

Solving nonlinear equations

$$f \in C^0([a, b])$$

$$\text{find } \xi \in [a, b] \text{ s.t. } f(\xi) = 0$$

↓
root or zero

$$f(a) \cdot f(b) < 0$$

Bisection algorithm

$$a_0 = a \quad b_0 = b \quad x_1 = \frac{a_0 + b_0}{2}$$

for $i = 1$ to N

$$\text{if } f(x_i) = 0 \text{ set } \xi = x_i \text{ stop}$$

$$\text{else if } f(x_i) \cdot f(a) < 0$$

$$a_i = a_{i-1} \quad b_i = x_i \quad x_{i+1} = \frac{a_i + b_i}{2}$$

$$\text{else if } f(x_i) \cdot f(b) < 0$$

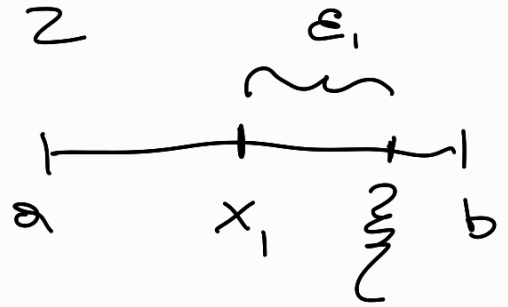
$$a_i = x_i \quad b_i = b_{i-1} \quad x_{i+1} = \frac{a_i + b_i}{2}$$

next i

$$\xi \in [a, b]$$

$$x_1 = \frac{a+b}{2}$$

$$\varepsilon_1 = |\xi - x_1| \leq \frac{b-a}{2}$$



$$L = b - a$$

$$L_i = 2^{-i} L$$

↑ tolerance

$$\varepsilon_i \leq \frac{1}{2} L_i \leq \tau$$

$$\frac{1}{2} L_i = 2^{-i} \frac{1}{2} L = 2^{-i-1} L \leq \tau$$

$$2^{-i-1} \leq \frac{\tau}{L} \quad \frac{L}{\tau} \leq 2^{i+1}$$

$$\log_2 \frac{L}{\tau} \leq i+1$$

$$\boxed{i \geq \log_2 \frac{L}{\tau} - 1}$$

"a priori" estimate

Convergence order

$$\{x_i\} \quad i=1, 2, 3, \dots \subset \mathbb{R}$$

$$x_i \xrightarrow{i \rightarrow \infty} \xi \quad x_i \text{ converges to } \xi$$

$$\lim_{i \rightarrow \infty} |\xi - x_i| = 0$$

• Linear convergence

$$\lim_{i \rightarrow \infty} \frac{|\xi - x_{i+1}|}{|\xi - x_i|} = \rho < 1$$

• Super-linear convergence

$$\lim_{i \rightarrow \infty} \frac{|\xi - x_{i+1}|}{|\xi - x_i|} = 0$$

• Convergence with order $p > 1$

$$\lim_{i \rightarrow \infty} \frac{|\xi - x_{i+1}|^p}{|\xi - x_i|^p} = c < \infty$$

Bisection converges with order $p = 1$

Newton's method

- Let $f \in C^2(\mathcal{U}(\xi))$
- $\mathcal{U}(\xi) \equiv (\xi - \delta, \xi + \delta)$
- $f(\xi) = 0$

$$f(x) = f(\xi) + f'(\xi)(x - \xi) + \mathcal{O}\left((x - \xi)^2\right)$$

$$\underline{\xi} \approx x - \frac{f(x)}{f'(\xi)} \approx x - \frac{f(x)}{f'(x)}$$

Newton-Raphson iteration

x_0 let $x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$

residual (pointing to $f(x_i)$)

increment (pointing to the subtraction term)

- Convergence depends on the initial guess
 . Whether the method converges or not depends on the choice of x_0
- Order of convergence depends on the function derivatives

Recast Newton-Raphson as a fixed point iteration

given $f \in C^0([a, b])$ find $\xi \in [a, b]$
 s.t. $f(\xi) = 0$

fixed point problem

$g \in C^0([a, b])$ $g([a, b]) \subseteq [a, b]$
 $\hookrightarrow$ fixed point map

find $\xi \in [a, b]$ s.t.

$$\boxed{g(\xi) = \xi}$$

fixed point

Consistent

$$g\left(\frac{\xi}{\eta}\right) = \frac{\xi}{\eta} \iff f\left(\frac{\xi}{\eta}\right) = 0$$

for example

$$g(x) = f(x) + x \quad \text{is consistent}$$

Fixed point iteration

given $x_0 \in [a, b]$

$$x_{i+1} = g(x_i) \quad i = 1, 2, \dots$$

Convergence

$$\lim_{i \rightarrow \infty} |x_i - \xi| = 0$$

Newton method

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$g_{\text{Newton}}(x) = x - \frac{f(x)}{f'(x)}$$

$$x_{i+1} = g_{\text{Newton}}(x_i)$$

consistent approximation
of Newton's method
if $f'(\xi) \neq 0$

$$f(\xi) = 0 \Rightarrow g(\xi) = \xi$$

Convergence? Order?

Theorem

$$g \in C^0([a, b])$$

$$g([a, b]) \subseteq [a, b]$$

$$\Rightarrow \exists \xi \in [a, b]$$

$$\text{s.t. } g(\xi) = \xi$$

Theorem

$$g \in C^0([a, b])$$

$$g([a, b]) \subseteq [a, b]$$

$$\exists L < 1 \text{ s.t. } \forall x, y \in [a, b] \quad |g(x) - g(y)| \leq L|x - y|$$

Lipschitz continuous

g is contractive

Then $\exists! \xi \in [a, b] \text{ s.t. } g(\xi) = \xi$

Theorem

$$\bullet g \in C^1([a, b])$$

$$\bullet g([a, b]) \subseteq [a, b]$$

$$\bullet \exists L < 1 \text{ s.t. } |g'(x)| \leq L \quad \forall x \in [a, b]$$

Then $\exists! \xi \in [a, b] \text{ s.t. } g(\xi) = \xi$

Theorem (4)

$$g \in C^0([a, b])$$

$$\exists \xi \in [a, b] \text{ s.t. } g(\xi) = \bar{g}$$

$$g \in C^1(\mathcal{U}(\xi))$$

$$|g'(\xi)| \leq L < \infty$$

Then $\exists \mathcal{U}'(\xi) \quad \mathcal{U}'(\xi) = (\xi - \delta, \xi + \delta)$

$$\exists! \xi \in \mathcal{U}'(\xi) \text{ s.t. } g(\xi) = \bar{g}$$

Theorem 2'

$$g \in C^0([a, b])$$

$$g([a, b]) \subseteq [a, b]$$

$$|g(x) - g(y)| \leq L |x - y|$$
$$L < 1$$

Then $\forall x_0 \in [a, b]$

$$x_{i+1} = g(x_i)$$

$$\lim_{i \rightarrow \infty} |x_i - \xi| = 0 \quad g(\xi) = \xi$$

Theorem 4' (Ostrowski)

$$g \in C^0([a, b])$$

$$g(\xi) = \xi \quad \xi \in [a, b]$$

$$g \in C^1(\mathcal{U}(\xi))$$

$$|g'(\xi)| \leq L < 1$$

Then $x_0 \in \mathcal{U}(\xi)$

$$x_{i+1} = g(x_i)$$

$$\lim_{i \rightarrow \infty} |x_i - \xi| = 0$$

$$\lim_{i \rightarrow \infty} \frac{|x_{i+1} - \xi|}{|x_i - \xi|} = g'(\xi)$$

Newton's method

$$g(x) = x - \frac{f(x)}{f'(x)}$$

$$g(\xi) = \xi$$

$$g'(x) = 1 - \frac{f'(x)f'(x) - f(x)f''(x)}{f'(x)^2}$$

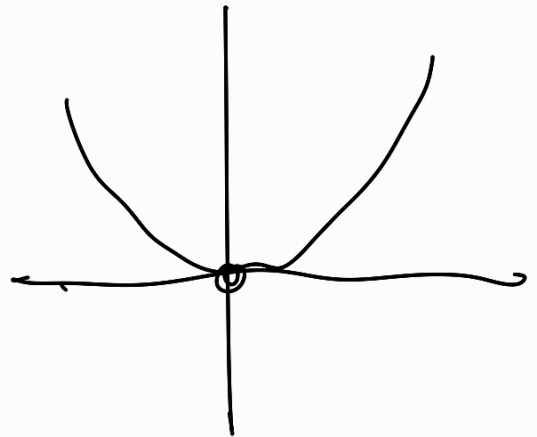
$$= \cancel{1} - \cancel{1} + \frac{f(x)f''(x)}{f'(x)^2}$$

$$g'(\xi) = \frac{f(\xi)f'(\xi)}{f'(\xi)} = 0$$

$$\boxed{f'(\xi) \neq 0} \Rightarrow \underline{\underline{g'(\xi) = 0}}$$

Example

$$f(x) = x^2$$



$$x_0 = 1$$

$$f'(x) = 2x$$

$$f'(x_0) = 2$$

$$x_1 = 1 - \frac{1}{2} = \frac{1}{2}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= \frac{1}{2} - \frac{1/4}{1} = \frac{1}{4}$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= \frac{1}{4} - \frac{1/16}{1/2} = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}$$

$$x_1 = \frac{1}{2} \quad x_2 = \frac{1}{4} \quad x_3 = \frac{1}{8}$$

Modified Newton method

Assume that ξ is a root of degree m of f

$$f(\xi) = f'(\xi) = f''(\xi) = \dots = f^{(m-1)}(\xi) = 0$$

Then the modified Newton's method

$$x_{i+1} = x_i - m \frac{f(x_i)}{f'(x_i)}$$

converges quadratically given a "good" initial guess x_0

$$f(x) = x^2$$

$$x_0 = 1$$

$$x_{i+1} = x_i - 2 \frac{f(x_i)}{f'(x_i)}$$

$$x_1 = 1 - \cancel{2} \frac{1}{\cancel{2} \cdot 1} = 0$$

$$= \infty$$

$$\lim_{i \rightarrow \infty} \frac{|x_{i+1} - \bar{x}|}{|x_i - \bar{x}|^p} = c$$

i large enough

$$\frac{|x_{i+1} - \bar{x}|}{|x_i - \bar{x}|^p} \approx c \approx \frac{|x_{i+2} - \bar{x}|}{|x_{i+1} - \bar{x}|^p}$$

$$\left(\frac{|x_{i+1} - \bar{x}|}{|x_i - \bar{x}|} \right)^p = \frac{|x_{i+2} - \bar{x}|}{|x_{i+1} - \bar{x}|}$$

$$p \approx \frac{\log \left(\frac{|x_{i+2} - \bar{x}|}{|x_{i+1} - \bar{x}|} \right)}{\log \left(\frac{|x_{i+1} - \bar{x}|}{|x_i - \bar{x}|} \right)}$$

e. posteriori estimate of p